Energy-Computing Synergy: How UHV Projects and Digital Dependence Empower Producer Services

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ABSTRACT

Overcoming energy supply constraints via cross-regional dispatch is vital for high-quality producer service development in the digital economy. Treating China's Ultra-High Voltage (UHV) transmission projects as a quasi-natural experiment, this study utilizes a multi-period Difference-in-Differences (DID) approach and 2008–2023 firm-level data to analyze the empowerment effects of new energy infrastructure. Findings indicate that UHV projects yield a "targeted empowerment" dividend rather than an economy-wide inclusive effect, specifically enhancing the performance of producer service firms in power-receiving regions (East). Mechanism analysis shows that digital dependence plays a critical moderating role; by ensuring a steady-state power supply, UHV projects reduce the Value of Lost Load (VoLL) and stimulate digital asset investment. Crucially, this empowerment is driven by power quality resilience rather than mere scale expansion of supply. Supporting evidence from 189,822 manually extracted digital asset records reveals a "diminishing marginal returns" pattern, where UHV infrastructure acts as an "equalizer" that benefits firms in digital transition, thereby narrowing the digital divide. Extended analysis demonstrates that this structural dividend substantiates the strategic synergy between the "West-to-East Power Transmission" and "Eastern Data and Western Computing" initiatives. These findings provide an empirical foundation for integrating energy-computing networks and constructing high-resilience energy systems in the AIGC era

1. Introduction

As the digital economy reshapes the global competitive landscape, the explosion of generative artificial intelligence (AIGC), represented by ChatGPT and Sora, has caused the dependence of computing power—as a core productive force—on energy systems to escalate exponentially. Sam Altman, the founder of OpenAI, pointed out that "the end of computing power is electricity." Distinct from traditional industries, the energy demand of producer services, such as data centers and cloud computing, has shifted from "total quantity abundance" to "power quality" guarantee. The International Energy Agency (IEA) predicts that global data center electricity consumption will double by 2026 (**?**); simultaneously, authoritative reports indicate that microsecond-level power interruptions have become the primary cause of data center outages (**?**). Globally, how to construct a power system with high resilience and high dispatching capacity to support large-scale computing power clusters has emerged as a strategic challenge universally faced by all nations.

From the perspective of macroeconomic structure (as illustrated in Figure **??**), China is undergoing a profound transformation. Since 2013, the growth rate of the value-added of the tertiary sector has consistently outpaced that of the secondary sector, becoming the main engine of growth. This transition is not solely an adjustment in proportion, but a transformation of the production function. As stated by **?**, AI technology is reshaping the macroeconomic production function, driving producer services to shift from labor input to digital capital deepening. In 2024, the value-added of the core industries of China's digital economy reached 10.5% of GDP (**?**). Against this backdrop, the energy demand of digital infrastructures such as data centers is escalating rapidly, with its electricity consumption share projected to rise to approximately 3.7% by 2030.

Figure **??** demonstrates that the node of large-scale construction of Ultra-High Voltage (UHV) transmission projects in 2014 highly coincides with the accelerated development of producer services. To address the structural contradiction of spatial mismatch between energy resources and load centers, China has constructed a systematic synergy of cross-regional dispatch through UHV transmission projects. This design of a unified national grid physically

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breaks through the constraints of China's Hu Huanyong Line on energy allocation. By delivering a steady-state power supply with high quality and minimal fluctuation, it satisfies the stringent requirements of eastern computing power clusters for latency and power quality, thereby providing a crucial steady-state energy guarantee for the high-end transformation of producer services.

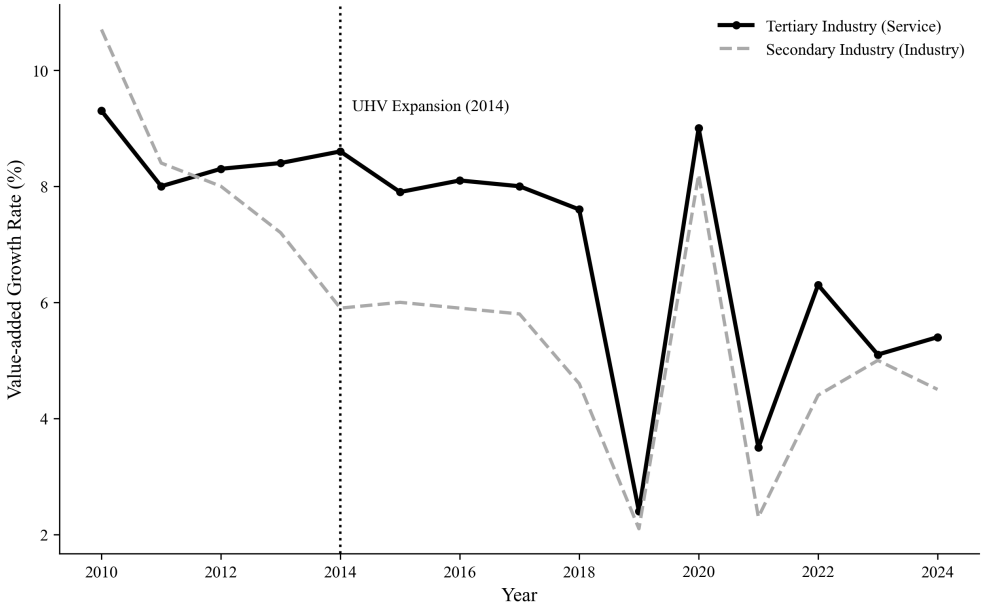

Figure 1: Temporal characteristics of China's industrial structure transformation and the spatiotemporal matching with UHV transmission project construction progress

Nevertheless, China faces a severe challenge of spatial mismatch in power supply and demand. As illustrated in Figure **??**, this study utilizes ArcGIS to map the comparison between the spatial distribution of net power surplus and producer services. Figure **??**(a) demonstrates the spatial characteristics of the energy supply side: benefiting from resource endowments, the western and northern regions have generated a significant power production surplus, indicated by the concentration of dark-colored areas. In stark contrast, Figure **??**(b) displays the spatial pattern of the energy demand side, where high value-added producer services are highly concentrated in the eastern coastal areas. This structural contradiction of production and consumption misalignment determines that the release of energy dividends must rely on highly efficient cross-regional transmission corridors.

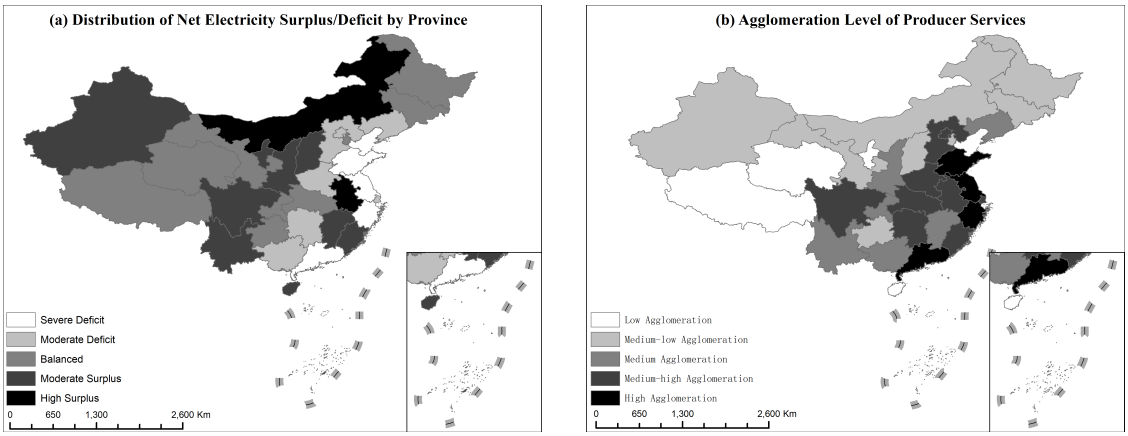

Figure 2: Comparison of spatial mismatch in China's power supply and demand and cross-regional allocation of UHV transmission projects

Note: Figure (a) illustrates the distribution of net power surplus across provinces, where dark colors represent surplus and light colors represent deficit; Figure (b) demonstrates the spatial distribution of the output value of producer services. The map classification utilizes the Jenks natural breaks optimization method to objectively reflect the structural differences of the indicators. Data are calculated based on the China Electric Power Statistical Yearbook 2024 and the China Statistical Yearbook 2024.

Consequently, China has implemented the West-to-East Power Transmission Project with Ultra-High Voltage (UHV) transmission projects as the backbone, aiming to overcome geographical barriers and achieve the precise coupling of western clean power generation capacity with eastern computing power load demands.

Against this backdrop, with the comprehensive launch of China's "Eastern Data and Western Computing" initiative, a critical strategic question emerges regarding whether there remains a strategic necessity for the eastern region to promote the West-to-East Power Transmission Project on a large scale, given that computing power hubs are shifting westward to locally consume green electricity. This study argues that the two are not substitutable in spatial layout, but rather exhibit deep synergy based on the characteristics of data elements. As pointed out by **?**, the key to constructing the integration of energy and computing networks lies in breaking the industry barriers between the energy network and the computing network. According to the strategic planning of China's National Development and Reform Commission (NDRC) (**?**), western computing power hubs primarily undertake "cold" data businesses with low requirements for network latency; conversely, "hot" data businesses—encompassing high-frequency financial transactions, real-time interactions, and corporate headquarters functions—are necessarily localized in the eastern region due to the physical latency constraints of optical fiber transmission. The continuous and high-intensity operation of these high-value businesses retained in the east is highly dependent on the steady-state energy guarantee delivered by cross-regional UHV transmission corridors. Therefore, UHV transmission projects essentially constitute a physical carrier bridging the western energy advantages with the core digital economy demands of the east, providing an indispensable resilience guarantee of energy for the digitalized operations of high-end producer services in the eastern region.

Extensive discussions have been conducted in the existing literature regarding the economic effects of UHV transmission projects. Early studies primarily analyzed their impacts on regional economic growth and carbon emissions from a macro perspective. Although discussions from a micro perspective are gradually increasing—for instance, **?** confirmed, based on manufacturing samples, that UHV projects significantly enhance firm production performance by alleviating regional power shortages and reducing energy consumption costs—this study posits that such traditional logic based on "total supply satisfaction" and "cost reduction" cannot fully explicate the empowerment mechanism of UHV transmission projects on modern services. In the context of the digital economy, the electricity demand of producer services, particularly fintech, internet, and artificial intelligence firms, has undergone a fundamental shift from reliance on "quantity" to stringent requirements for power supply reliability and power quality. For firms highly dependent on digital technology, power stability is the core prerequisite for ensuring their "Business Continuity" (**?**). Research by authoritative institutions indicates that power failure is the primary cause of service interruptions in data centers, with the average economic loss of a single unplanned outage reaching approximately 505,000 USD (**?**), and that high-frequency trading and AI model training are exceedingly sensitive to voltage sags and harmonic interference (**?**). However, existing studies have yet to deeply investigate, from the heterogeneous perspective

of "digital dependence," how UHV projects structurally enhance the performance of high-tech services by providing a steady-state power supply to circumvent the risks of data loss and business interruption.

In view of this, this study treats the commissioning of UHV transmission projects as a quasi-natural experiment. Based on the data of Chinese A-share listed service firms from 2008 to 2023, a multi-period Difference-in-Differences (DID) model is employed to systematically examine the impact and functional mechanisms of energy infrastructure on the production performance of producer services. The findings reveal that: First, contrary to the conclusion of universal benefits in the manufacturing sector, the enhancement effect of UHV projects on the performance of the full service industry sample is statistically insignificant, indicating that a mere "total quantity supply" does not yield an inclusive dividend. Second, structural analysis reveals a significant "precision irrigation" characteristic (i.e., targeted empowerment). After focusing on the core subsample of "eastern power-receiving end × modern services," UHV projects exhibit a significant enhancement effect on firm performance (with a coefficient of 0.082), verifying their targeted support capability for high-tech industries. Third, mechanism testing demonstrates that the degree of digital dependence plays a crucial positive moderating role. That is, the higher the degree of digitalization of a firm, the greater its risk exposure to power interruptions (measured by VOLL), and consequently, the greater the marginal steady-state dividend it can derive from the steady-state power supply provided by UHV projects. This provides micro-level evidence for the complementarity of "energy-digital" elements.

The marginal contributions of this study are threefold. First, it advances the research frontier regarding the economic effects of energy infrastructure by pivoting the analytical focus from cost substitution" to production continuity guarantee." Unlike prior literature that predominantly examines cost-reduction effects in the manufacturing sector, this study pioneers the perspective of producer services. It reveals how UHV projects—by delivering a steady-state power supply—mitigate the tail risks faced by high-tech producer service firms, thereby safeguarding business continuity. This elucidates why specific service subsectors can capture excess returns even when the full-sample effect remains statistically insignificant. Second, it uncovers the micro-level moderating mechanism through which UHV infrastructure shapes the digital economy. Leveraging the large language model-based firm-level digitalization measurement framework developed by **?**, this study demonstrates a significant configuration effect between UHV projects and corporate digital transformation. Firms with higher levels of digitalization exhibit greater sensitivity to power supply quality, revealing how energy infrastructure converts physical supply" into a productive input" for the digital economy by satisfying high-tech firms' rigid demands for a steady-state power supply. Third, it provides spatially grounded empirical evidence for the energy-computing-data" integration strategy. The findings rigorously validate the internal logical consistency between the West-to-East Power Transmission" and Eastern Data and Western Computing" national strategies. They demonstrate how UHV cross-regionally converts western energy endowments into eastern computational advantages and alleviates the energy bottlenecks constraining local hot data" processing, yielding important policy implications for optimizing China's digital infrastructure layout.

Finally, although the sample period of this study concludes in 2023, the findings offer critical foresight for the current AIGC (Generative AI) era. The training and inference of large AI models impose unprecedented demands on the stability and green-energy intensity of data centers. The "energy-digital coupling" mechanism unveiled in this paper suggests that UHV infrastructure is not merely a driver for past service sector transitions, but also a strategic cornerstone for cross-regional energy-computing synergy in the impending Artificial Intelligence revolution. This study thus provides a timely empirical foundation for understanding how infrastructure layout determines the ceiling of a nation's digital productivity.

The remainder of this study is structured as follows: Section 2 analyzes the institutional background of UHV transmission projects and proposes theoretical hypotheses combined with the characteristics of the digital economy; Section 3 delineates the research design, including sample selection, model specification, and variable measurement; Section 4 reports the results of baseline regressions, the parallel trend test, and a series of robustness checks; Section 5 conducts a mechanism analysis from a micro perspective to verify the moderating role of digital dependence and distinguish between the "quality" and "quantity" attributes of the empowerment dividend; Section 6 further explores the spatial heterogeneity of the empowerment effect and its synergistic logic with the "Eastern Data and Western Computing" initiative from a macro perspective; and Section 7 concludes the study and puts forward policy recommendations.

2. Institutional Background and Theoretical Mechanisms

2.1. Institutional Background of UHV Transmission Project Construction

There is a significant spatial mismatch between the supply and demand sides of China's power system. This structural contradiction has transcended the traditional scope of "natural resource endowment" and evolved into a systemic imbalance between "generation capacity" and "load centers." As pointed out by **?**, driven by the new energy security strategy, China has constructed the world's largest clean energy power generation bases in the western and northern regions, forming a massive power surplus; however, producer services and digital infrastructure, which account for more than 70% of the country's electricity consumption demand, are highly concentrated in the eastern and central regions, leading to a long-term power deficit in these areas. This characteristic of "inverse spatial distribution of energy production and consumption" causes traditional Extra-High Voltage (EHV, e.g., 500kV) transmission systems to face technical bottlenecks such as high transmission losses, scarcity of corridor resources, and poor system stability when transmitting power over long distances across thousands of kilometers. To resolve this structural contradiction that constrains national energy security and economic development, China first proposed the strategic concept of constructing Ultra-High Voltage (UHV) transmission projects in 2004, aiming to achieve the cross-regional, long-distance, large-capacity, and low-loss efficient dispatch of power resources by constructing transmission networks with voltage levels of 1000kV AC, ±800kV DC, and above.

In 2006, UHV transmission technology was formally incorporated into the *National Medium- and Long-Term Scientific and Technological Development Plan Outline*. In 2009, China's first 1000kV UHV AC test and demonstration project (Jindongnan–Nanyang–Jingmen) was put into commercial operation; in 2010, the Xiangjiaba–Shanghai ±800kV UHV DC project was commissioned, marking the official entry of the strategy of the West-to-East Power Transmission Project into the UHV era. Notably, 2014 served as a critical turning point for UHV construction. To address the pressures of air pollution prevention and control and the energy structure transition in the eastern region, China's National Energy Administration (NEA) issued the *Notice on Accelerating the Construction of 12 Key Transmission Corridors for the Air Pollution Prevention and Control Action Plan* (**?**), and UHV projects entered a phase of regular construction. As of October 2025, during the "14th Five-Year Plan" period, China's UHV project construction has progressed steadily, with a total of 45 UHV projects commissioned, and the cross-province and cross-region transmission capacity of the West-to-East Power Transmission Project is expected to exceed 340 million kilowatts. This interconnected power grid architecture covering the entire country constitutes the physical cornerstone supporting the high-quality development of China's digital economy.

The typical construction paths and commissioning times of UHV projects are presented in Table **??**.

Table 1: Typical construction paths and commissioning times of UHV transmission projects

Commissioning Date	UHV Transmission Line Name	Commissioned Province (Converter Station)
2010.06	Yunnan–Guangzhou ±800kV UHV DC Transmission Project	**Guangdong**
2010.07	Xiangjiaba–Shanghai ±800kV UHV DC Transmission Project	**Shanghai**
2012.12	Jinping–Sunan ±800kV UHV DC Transmission Project	**Jiangsu**
2013.09	Huainan–Zhebei–Shanghai 1000kV UHV AC Transmission Project	**Zhejiang, Shanghai**
2013.09	Pu'er–Jiangmen ±800kV UHV DC Transmission Project	**Guangdong**
2014.06	Nuozhadu–Guangdong ±800kV UHV DC Transmission Project	**Guangdong**
2014.07	Xiluodu–Zhexi ±800kV UHV DC Transmission Project	**Zhejiang**
2014.12	Zhebei–Fuzhou 1000kV UHV AC Transmission Project	**Zhejiang, Fujian**

Note: The table is compiled based on data from the State Grid Corporation of China and relevant statistical yearbooks. Considering the sample observation period and the requirements for identifying policy shocks in this study, the table primarily lists representative transmission lines from the initial phase of construction.

It is worth emphasizing that the construction of Ultra-High Voltage (UHV) transmission projects exhibits typical exogenous characteristics, providing a quasi-natural experimental foundation for the application of the multi-period Difference-in-Differences (DID) method in this study. First, UHV projects are national-level major energy

infrastructures. Their planning and construction are centrally coordinated by the State Grid Corporation of China according to national macro energy strategies and implemented following approval from the central government. This implies that individual firms or even single cities lack the capacity to influence the routing and terminal points of UHV lines through their own lobbying capabilities. Second, UHV projects are characterized by long construction cycles and high technical thresholds, typically requiring 3 to 5 years from initiation to commissioning. Micro-level firms are unable to accurately predict specific commissioning timelines to adjust their production decisions in advance. Therefore, the operation of UHV projects can be regarded as a typical "exogenous shock" for producer service firms.

2.2. Technical Features of UHV: From "Total Supply" to "Resilience Reshaping"

Compared with traditional power grids, UHV projects represent not only a doubling of transmission capacity but also a fundamental transformation of the power supply mode from mere "overall supply-demand balance" to "high-quality reliable supply" (**?**).

First, in the dimension of total supply, the transmission power of UHV DC lines can reach over 8 million kilowatts, enabling the large-scale transmission of surplus clean energy from the west to eastern load centers. As **?** noted, digital new infrastructure exhibits significant "distance decay characteristics," whereas UHV projects directly break through regional energy barriers through physical-level ultra-long-distance transmission, effectively alleviating the power supply-demand gap caused by the scarcity of primary energy in receiving areas.

Second, in the dimension of resilience reshaping, UHV grids integrate advanced Voltage Source Converter-based High Voltage Direct Current (VSC-HVDC) technology and intelligent dispatch systems, which are particularly critical for the digital economy (**?**). Compared with traditional grids, UHV systems possess millisecond-level active/reactive power regulation capabilities and superior fault ride-through capabilities, effectively smoothing voltage sags and harmonic interference triggered by new energy integration or load fluctuations. This generational technological leap renders UHV projects not only energy transmission corridors but also the physical foundation for constructing high-reliability and high-resilience new power systems. They provide a steady-state power supply with pure waveforms and minimal interruptions for downstream digital services (e.g., data centers and financial trading systems) that are extremely sensitive to power quality. Notably, this technical trajectory has become a core focus of national energy strategy. According to the "15th Five-Year Plan" released by the State Grid, 4 trillion RMB will be invested over the next five years to build a new power system. It explicitly proposes the comprehensive promotion of VSC-HVDC projects from the Qinghai-Tibet Plateau to core load centers like the Guangdong-Hong Kong-Macao Greater Bay Area to fundamentally resolve stability challenges at the physical level (**?**).

2.3. Theoretical Analysis and Research Hypotheses

With the deep integration of the digital economy and the real economy, the production function of modern services has undergone structural changes. According to the Directed Technical Change theory, the efficiency release of high-tech capital often depends on the synergistic effect of specific complementary elements (**?**). In modern services, "digital capital"—represented by data centers, high-performance servers, and complex algorithms—has ascended to become a core production factor, and its effectiveness is highly dependent on the "steady-state" and "greening" of the power supply. UHV transmission projects empower the performance of service firms through the reshaping of power supply modes across the following three dimensions:

1. Supply-side shocks and industry heterogeneity: Precise identification based on the "dilution effect":

The impact of infrastructure construction on micro-level firms exhibits significant industry selectivity. Traditional consumer services (e.g., catering and accommodation) are labor-intensive industries, where electricity accounts for a negligible portion of the total cost structure. According to the law of diminishing marginal returns of factors, the marginal contribution of a mere increase in power supply to such discontinuous production, low-electricity-dependent firms is extremely limited. Therefore, from an industry-wide perspective, the presence of a large number of traditional service firms will generate a strong "average effect dilution," making policy dividends statistically difficult to identify. In contrast, modern services (information transmission, finance, and scientific research) have achieved high levels of digital capital deepening. Their business logic is based on the real-time processing and exchange of massive data, where electricity is not only an energy input but also the "basic environment" for maintaining the operation of digital capital. UHV projects resolve the "hard power shortage" constraints of eastern load centers through large-capacity, cross-regional power dispatch. This empowerment effect exhibits distinct "precision irrigation" characteristics (i.e., targeted empowerment)—specifically, driving the structural upgrading of the service industry by removing the energy weaknesses of specific high-productivity industries, rather than through inclusive cost subsidies.

2. Moderating effects of digital dependence: Risk avoidance and investment incentive mechanisms:

The degree of digital dependence is not only a technical characteristic of a firm but also an "amplifier" of its sensitivity to energy stability. Firms with a higher degree of digitalization are better able to achieve performance leaps by utilizing the steady-state dividends brought by UHV projects.

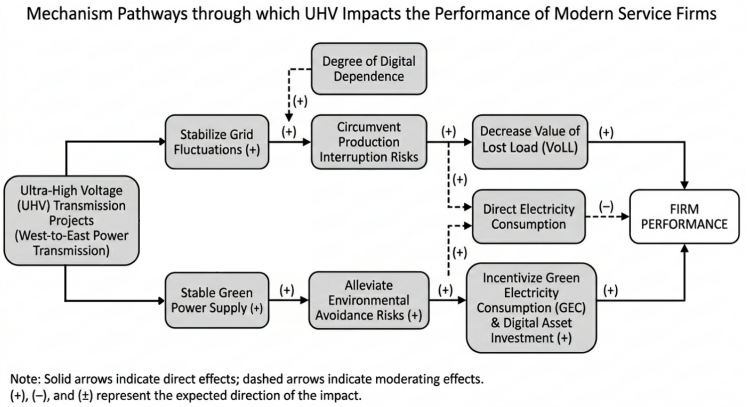

Figure 3: **Mechanism pathways through which UHV transmission projects impact the performance of modern service firms**

(1) Passive constraint perspective: Value of Lost Load (VoLL) and business continuity guarantee.

In energy economics, the Value of Lost Load (VoLL) is a core metric for measuring power reliability. Digital infrastructure exhibits extreme sensitivity to power quality. For highly digitalized service firms, such as those involved in generative AI R&D and high-frequency financial trading, the production process is characterized by "strong continuity and irreversibility." Even millisecond-level voltage sags or harmonic interference can trigger server downtime, massive data corruption, or the interruption of core trading logic. The hidden opportunity costs resulting from such incidents, such as customer churn and system reconstruction, far exceed explicit energy costs. By stabilizing grid fluctuations through advanced VSC-HVDC technology, Ultra-High Voltage (UHV) transmission projects provide an "energy safety cushion" for highly digitalized firms, enhancing firm performance from the passive constraint side by circumventing production interruption risks.

(2) Active empowerment perspective: Green electricity consumption (GEC) and digital asset investment incentives.

From the perspective of active allocation, digital firms face heavy pressure regarding green electricity consumption (GEC) under the context of China's "dual-carbon" strategies and stringent Power Usage Effectiveness (PUE) constraints. As the green artery of the West-to-East Power Transmission Project, UHV transmission projects deliver electricity containing a high proportion of clean energy from western wind and solar sources. This expectation of long-term, stable green power supply effectively alleviates "environmental avoidance risks" in the digital transformation of firms, thereby generating an investment incentive effect: encouraging firms to proactively expand capital expenditures on digital assets such as computing clusters and cloud platforms, driving performance enhancement through economies of scale and factor synergy.

3. Spatial synergy effect: Complementary logic based on the "integration of energy, computing, and networks":

The economic effects of UHV projects also depend on the spatial matching efficiency between energy supply and computing power demand. As per the "integration of energy and computing networks" framework proposed by **?**, the effectiveness of national data infrastructure relies on the deep fusion of energy and computing networks. UHV transmission projects serve not only as physical channels for energy but also as the underlying support environment for the trans-regional dispatch of computing resources and the efficient circulation of data elements. Under the strategic landscape of the "Eastern Data and Western Computing" initiative (**?**), China's digital economy exhibits a distinct functional division: the West primarily handles "cold" data businesses that are insensitive to network latency (e.g., storage and backup), while the East aggregates "hot" data businesses that are extremely latency-sensitive, such as financial trading, real-time interaction, and the industrial internet. Due to the insurmountable communication latency caused by physical distance, eastern "hot data" industries cannot migrate westward on a large scale; their continuous

operation is highly dependent on trans-regional "long-distance power transmission." UHV essentially constitutes a physical tie connecting "western energy advantages" with "eastern high-value digital demand," achieving strategic complementarity where "hot data is computed in the East while electricity comes from the West." In contrast, although the western region has abundant power, the micro-conversion efficiency of its energy steady-state dividend is lower due to the lack of high-value-added digital application scenarios.

Based on the above analysis, this study proposes:

Research Hypothesis 1: The empowerment effect of UHV transmission projects on the service industry does not manifest as an inclusive "total supply replenishment," but rather presents significant structural characteristics, specifically by precisely enhancing the operational performance of producer service firms located at the power-receiving end (East).

Research Hypothesis 2: The degree of digital dependence plays a positive moderating role in the impact of UHV on firm performance: that is, the higher the degree of digitalization, the higher the firm's Value of Lost Load, and the greater the marginal performance steady-state dividends it derives from the steady-state power supply and "green power" provided by UHV projects.

Research Hypothesis 3: Deep synergy exists between UHV projects and the "Eastern Data and Western Computing" initiative. Its strategic value lies in forming an energy resilience foundation supporting the development of "hot data" industries in the eastern region, with the empowerment effect exhibiting significant spatial non-equilibrium.

3. Research Design

3.1. Data Source and Sample Selection

This study utilizes Chinese A-share listed service firms from 2008 to 2023 as the initial research sample. The selection of this sample period is primarily based on the following considerations: 2008 marks the beginning of the substantive operation phase of China's Ultra-High Voltage (UHV) transmission projects, providing a starting point that covers the complete cycle of the UHV network from pilot demonstrations to large-scale regular construction. Furthermore, data up to 2023 can capture the latest changes in the demand for energy resilience in the service sector against the backdrop of the generative artificial intelligence explosion.

Micro-level firm financial and corporate governance data required for this study are primarily sourced from the CSMAR and Wind databases. Information regarding UHV project names, commissioning dates, and terminal provinces was manually collected and organized through announcements from China's National Energy Administration (NEA) and annual reports of the State Grid Corporation of China (SGCC). To ensure the validity and precision of the empirical estimations, this study conducted rigorous "structural identification" and screening of the initial sample according to the *Statistical Classification of Producer Services (2019)* standards issued by China's National Bureau of Statistics (**?**):

Precise industry definition: Diverging from the generalized treatment of the service sector in previous literature, this study strictly limits the research objects to producer services that "provide supporting services for production activities." Specifically, the following core industry categories are retained: Transportation, Storage, and Post (G); Information Transmission, Software, and Information Technology Services (I); Financial Intermediation (J); Leasing and Business Services (L); Scientific Research and Technical Services (M); and Wholesale (F). Concurrently, samples from consumer services such as accommodation and catering, residential services, and education and health are excluded. The rationale for this screening is that the production function of consumer services is highly dependent on labor input, with electricity costs accounting for a negligible proportion and low sensitivity to power quality; including the full sample would lead to severe "average effect dilution" in statistical terms, obscuring the authentic empowerment dividends of the infrastructure.

Data cleaning: Firm samples with abnormal financial status, marked as ST, *ST, or PT during the sample period, are excluded. Observations with missing or negative values for key financial data (e.g., operating revenue, total assets, and number of employees) are also removed.

Winsorization: To mitigate the potential interference of extreme outliers on the regression results, all continuous variables are Winsorized at the 1% and 99% quantiles.

Following the aforementioned cleaning procedures, this study ultimately obtained unbalanced panel data covering 14,358 firm-year observations.

3.2. Econometric Model Specification

Following the empirical strategy of **?**, this study treats the commissioning of UHV projects as a quasi-natural experiment and constructs a multi-period Difference-in-Differences (DID) model to examine the impact of energy supply security on the performance of service firms:

$$Y_{it} = \alpha + \beta \cdot \mathrm{UHV}_{it} + \gamma \cdot X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{1}$$

Where the subscripts i and t represent the firm and year, respectively. The dependent variable Y_{it} is the production performance of firm i in year t. The core independent variable UHV_{it} is the UHV shock dummy variable. β reflects the average treatment effect of UHV project commissioning on the performance of service firms. X_{it} represents the set of firm-level control variables, μ_i is the firm individual fixed effect, λ_t is the year fixed effect, and ε_{it} is the random disturbance term.

Furthermore, to verify the moderating role of the degree of digital dependence, this study introduces a digitalization index ($\mathrm{Digital}_{it}$) and its interaction term with UHV_{it} based on Model (**??**) to construct a moderation effect model:

$$Y_{it} = \alpha + \beta_1 \cdot \mathrm{UHV}_{it} + \beta_2 \cdot \mathrm{Digital}_{it} + \beta_3 \cdot \mathrm{UHV}_{it} \times \mathrm{Digital}_{it} + \gamma \cdot X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{2}$$

In Model (**??**), the interaction term coefficient β_3 is the focus of this study. If it is significantly positive, it validates the "digital bias" characteristic of the UHV empowerment effect.

3.3. Variable Selection and Measurement

1. Dependent variable: This study selects the natural logarithm of operating revenue (ln(Sales)) as the core metric to measure firm production performance. This indicator intuitively reflects the market scale and operational level of a firm and is relatively less affected by accounting measurement policies. In robustness checks, Return on Assets (ROA) is further adopted as a supplementary indicator.

2. Core independent variable: The core independent variable is the UHV project shock (UHV_{it}), defined as the interaction between the treatment group dummy variable Treat_i and the time dummy variable Post_t. Specifically, $\mathrm{Treat}_i = 1$ if the province where the firm is registered has established a UHV converter station, and 0 otherwise. If it is the year of commissioning of the first UHV line in that province or subsequent years, $\mathrm{Post}_t = 1$, and 0 otherwise.

3. Moderating variable: The core moderating variable of this study is the firm's degree of digital dependence ($\mathrm{Digital}_{it}$). To overcome the defects of traditional single indicators (e.g., whether there is an official website or whether software is purchased) in measuring the depth of firm digitalization, this study introduces and applies the firm digitalization measurement framework constructed by **?** based on large language models. The specific measurement steps are as follows:

Corpus construction: Based on authoritative literature, 78 core keywords covering five major dimensions—"Artificial Intelligence, Big Data, Cloud Computing, Blockchain, and Digital Technology Application"—were screened (e.g., algorithms, cloud services, data mining, distributed computing, etc.).

Text extraction: Utilizing Python text analysis technology, the "Management Discussion and Analysis (MD&A)" section in the annual reports of the sample firms was subjected to full-text crawling and cleaning.

Indicator synthesis: The total frequency of the aforementioned keywords appearing in each annual report was counted. Considering the right-skewed distribution of word frequency data, this study follows standard practice by adding 1 to the total word frequency and then taking the natural logarithm, i.e., $\mathrm{Digital}_{it} = \ln(\text{Total Frequency}_{it} + 1)$. A higher value of this indicator indicates a deeper application of digital technology in the firm's production and operation, thereby generating a stronger "technical dependence" on the continuity and stability of power supply.

To complement the text-based digitalization measure and ensure robustness, this study further constructs an objective indicator of substantive digital investment ($\mathrm{Digital_Obj}_{it}$). We retrieved the "Intangible Asset Details" from the CSMAR database for the period 2008–2023. A Python-based keyword matching algorithm was applied to the "Subject Category" and "Description" fields using keywords such as "software," "system," "network," "platform," "data," "ERP," "website," "cloud," "management information," and "APP." This exhaustive process successfully identified and extracted 189,822 valid digital asset records. We then aggregated these records by firm and year to

calculate the total year-end balance of digital intangible assets. Digital_Obj_{it} is defined as the ratio of these digital-related intangible assets to the firm's total assets. This objective measure captures the firm's substantive commitment to digitalization, providing a parallel validation for the strategic intent captured by the text analysis.

4. Control variables: Given data availability and to maintain consistency with industry baseline studies, this study selects core financial indicators such as firm size, leverage ratio, and profitability as control variables, which cover the primary dimensions affecting firm performance. Specifically, these include: (1) Firm Size (Size), measured by the natural logarithm of total assets; (2) Leverage Ratio (Lev), reflecting financial leverage risk; (3) Listing Age (Age), measured by the natural logarithm of the difference between the observation year and the firm's listing year plus 1; (4) Return on Assets (ROA), measuring firm profitability; and (5) Operating Revenue Growth Rate (Growth), reflecting firm growth potential.

Specific variable definitions and descriptions are provided in Table **??**.

Table 2: Variable definitions and descriptions

Variable Type	**Symbol**	**Definition and Calculation**
Dependent Variables	ln(Sales)	Natural logarithm of operating revenue
	ROA	Net profit divided by the average balance of total assets
Core Independent Variable	UHV_{it}	UHV project shock dummy variable ($\text{Treat}_i \times \text{Post}_t$)
Moderating Variables	Digital_{it}	Strategic digital dependence: Logarithm of (total frequency of digitalization-related keywords in annual reports + 1)
	Digital_Obj_{it}	Substantive digital investment: Ratio of digital-related intangible assets to total assets
Control Variables	Size	Firm size: Natural logarithm of total assets
	Lev	Leverage: Total liabilities divided by total assets
	Age	Firm age: Logarithm of (observation year − founding year + 1)
	ROA	Return on Assets: Net profit divided by the average balance of total assets
	Growth	Growth rate: (Current operating revenue − previous revenue) / previous revenue

3.4. Descriptive Statistics

Table **??** reports the descriptive statistics of the main variables. The cleaned sample includes 14,358 firm-year observations. Regarding statistical characteristics, the mean operating revenue (logarithm) of the sample firms is 21.228, with a standard deviation of 1.404, showing a reasonable distribution. The mean of the UHV shock (DID) variable is 0.416, indicating that approximately 41.6% of the observations were affected by UHV projects. The mean degree of digital dependence is 2.139, with a standard deviation of 1.615, a minimum of 0, and a maximum of 4.997, showing significant variation and providing an adequate statistical basis for examining its moderating effect. The mean firm size (logarithm of total assets) is 22.295, the mean leverage ratio is 0.449, the mean return on assets is 0.022, the mean listing age (logarithm) is 1.576, and the mean operating revenue growth rate is 0.128; all variables fall within reasonable ranges.

Table 3: Descriptive statistics of main variables

Variable	Obs	Mean	Std. Dev.	Min	Max
Operating Revenue (ln)	14358	21.228	1.404	18.868	24.069
UHV Shock (DID)	14358	0.416	0.493	0.000	1.000
Digital Dependence	14358	2.139	1.615	0.000	4.997
Firm Size	14358	22.295	1.385	20.196	25.881
Leverage	14358	0.449	0.218	0.105	0.892
ROA	14358	0.022	0.055	-0.118	0.112
Listing Age (ln)	14358	1.576	0.808	0.693	3.219
Growth Rate	14358	0.128	0.481	-0.756	2.942

3.5. Correlation Analysis

Table **??** presents the Pearson correlation coefficient matrix for the main variables. The results show that the UHV shock (DID) is significantly positively correlated with operating revenue (logarithm) at the 1% level (correlation coefficient of 0.107), providing preliminary support for Research Hypothesis 1. Digital dependence also shows a significant positive correlation with operating revenue (logarithm) (correlation coefficient of 0.0947), suggesting that firms with a higher degree of digitalization typically possess better operational performance. The correlation coefficient between firm size and operating revenue (logarithm) is as high as 0.851, which aligns with economic intuition that larger firms usually have higher operating revenues. Leverage ratio is positively correlated with operating revenue (0.314) and negatively correlated with return on assets (−0.329), reflecting the dual effect of financial leverage. The correlation coefficients between all control variables do not exceed 0.5, indicating that multicollinearity is not a serious issue and providing a sound foundation for the subsequent regression analysis.

Table 4: Pearson correlation coefficient matrix of main variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) Operating Revenue	1							
(2) UHV Shock	0.107***	1						
(3) Digital Dependence	0.0947*	-0.0177	1					
(4) Firm Size	0.851*	0.128*	-0.0224***	1				
(5) Leverage	0.314*	0.0427*	-0.209*	0.375*	1			
(6) ROA	0.188*	-0.0181	-0.0870*	0.0901*	-0.329***	1		
(7) Listing Age	0.0859*	0.0815*	-0.0796*	0.220*	0.175*	-0.0678*	1	
(8) Growth Rate	0.0565*	-0.0125	-0.0945*	-0.000432	0.0306*	0.224*	0.0185**	1

Note: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

4. Analysis of Empirical Results

4.1. Baseline Regression Analysis: From Total Supply to Targeted Empowerment

Table **??** reports the baseline regression results regarding the impact of Ultra-High Voltage (UHV) transmission projects on the performance of service firms. To deeply analyze the structural characteristics of the policy effect, this study adopts an identification strategy that progressively narrows the sample scope: Column (1) uses the full sample of the service industry for estimation; Column (2) focuses on the subsample of modern services; and Column (3) further isolates the core sample of modern services located at the power-receiving end (East). All regressions control for firm individual fixed effects and year fixed effects, with standard errors clustered at the prefecture-level city level.

Table 5: Baseline regression results: Structural identification based on industry and region

	Dependent Variable: Firm Performance		
	(1) Full sample	(2) Producer services	(3) Eastern producer services
DID	0.017	0.067	0.082*
	(0.51)	(1.58)	(1.80)
Size	0.862***	0.843***	0.831***
	(63.06)	(39.85)	(36.82)
Lev	0.041	-0.025	-0.016
	(0.87)	(-0.36)	(-0.22)
LnAge	0.022	0.089***	0.091***
	(1.55)	(4.42)	(4.33)
cons	2.004***	2.271***	2.530***
	(6.76)	(5.00)	(5.23)
Observations	19608	5857	5568
Adjusted R^2	0.933	0.925	0.925

Note: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. t-statistics are reported in parentheses. All regressions include firm fixed effects and year fixed effects, with standard errors clustered at the prefecture-city level.

According to the full-sample regression results in Column (1), the coefficient of the core independent variable DID is 0.017 and is statistically insignificant ($t = 0.51$). This result indicates that without considering the differences in industry attributes and regional functions, a mere increase in power supply does not lead to an inclusive performance enhancement at the overall service industry level. This confirms the previous theoretical analysis: a large number of traditional consumer services at the low end of the industrial chain have low sensitivity to power factors, which statistically dilutes the average treatment effect of the policy.

Column (2) excludes the traditional service samples, retaining only producer service firms. The results show that the coefficient of DID increases to 0.067, and the t-value rises to 1.58. Although this approaches the 10% significance level, it does not fully cross the threshold of statistical significance. This suggests that considering industry attributes alone is insufficient to fully capture the economic dividends of UHV; the matching of regional energy supply and demand is equally critical.

Column (3) further presents the regression results for the core subsample of "Eastern Receiving End × Producer Services". In this specification, the coefficient of DID further increases to 0.082 and is significant at the 10% level ($t = 1.80$). From an economic perspective, this means that after the commissioning of UHV transmission projects, the operating revenue of modern service firms located in eastern load centers increased by approximately 8.2% on average. Regarding the control variables, firm size (Size) and listing age (LnAge) are both significantly and positively correlated with firm performance, aligning with general economic intuition.

The aforementioned progressive empirical results strongly support Research Hypothesis H1. They reveal that the economic effect of UHV projects does not manifest as a "blanket" supply replenishment, but rather exhibits distinct "precision irrigation" characteristics (i.e., targeted empowerment): the productivity dividends of the infrastructure can only be effectively released when "high-quality energy supply" (UHV) and "high energy-sensitivity demand" (eastern modern services) are precisely coupled in space.

4.2. Identification Condition Test: Parallel Trends and Dynamic Effects

The prerequisite for evaluating policy effects using the Difference-in-Differences (DID) method is satisfying the parallel trend assumption, which requires that treatment and control group firms maintain the same development trend prior to policy implementation. To verify that the impact of UHV transmission projects on producer services satisfies this assumption and to examine the dynamic characteristics of the policy effect, this study adopts the event study approach following Jacobson et al. (1993) to construct the following test model:

$$\text{LnRevenue}_{it} = \beta_0 + \sum_{k=-4,k\neq-1}^{4} \beta_k \text{DID}_{it}^k + \gamma X_{it} + \mu_i + \delta_t + \epsilon_{it} \tag{3}$$

where DID_{it}^{k} represents a series of dummy variables reflecting the years relative to the commissioning of UHV projects. Specifically, assuming c_i is the year the first UHV line was officially commissioned in the province where firm i is located, if $t - c_i = k$, then $\mathrm{DID}_{it}^{k} = 1$, otherwise 0. Considering the identification characteristics of multi-period DID, the observation window is set from 4 years before to 4 years after the policy occurrence, with the period immediately preceding the policy ($k = -1$) as the baseline period. Other variables in the model are set consistently with the baseline regression model. The coefficient β_k reflects the changes in performance differences between the treatment and control groups in a specific relative year.

Figure **??** illustrates the dynamic effect test results at a 95% confidence level, using operating revenue and Return on Assets (ROA) as the dependent variables. It can be observed that before the commissioning of UHV projects ($k < 0$), the estimated coefficients for each period fluctuate slightly around zero, and their confidence intervals all contain zero, making them statistically insignificant. This indicates that prior to the policy shock, the development paths of firms in the treatment and control groups were essentially parallel, with no significant pre-existing differences, thus the DID model designed in this study passes the parallel trend test.

Regarding the dynamic trend after policy commissioning ($k \geq 0$), the estimated coefficients for both operating revenue and profitability (ROA) shift significantly upward from near zero and generally remain within the positive range. This demonstrates that the steady-state dividend of the steady-state power supply provided by UHV projects effectively translated into scale expansion and efficiency improvements for producer service firms. Although some long-term coefficients have slightly wider confidence intervals due to statistical power constraints after sample splitting for single years, the overall trend maintains high consistency with the previous baseline regression conclusions, reflecting the dynamic resilience of the empowerment effect of new energy infrastructure.

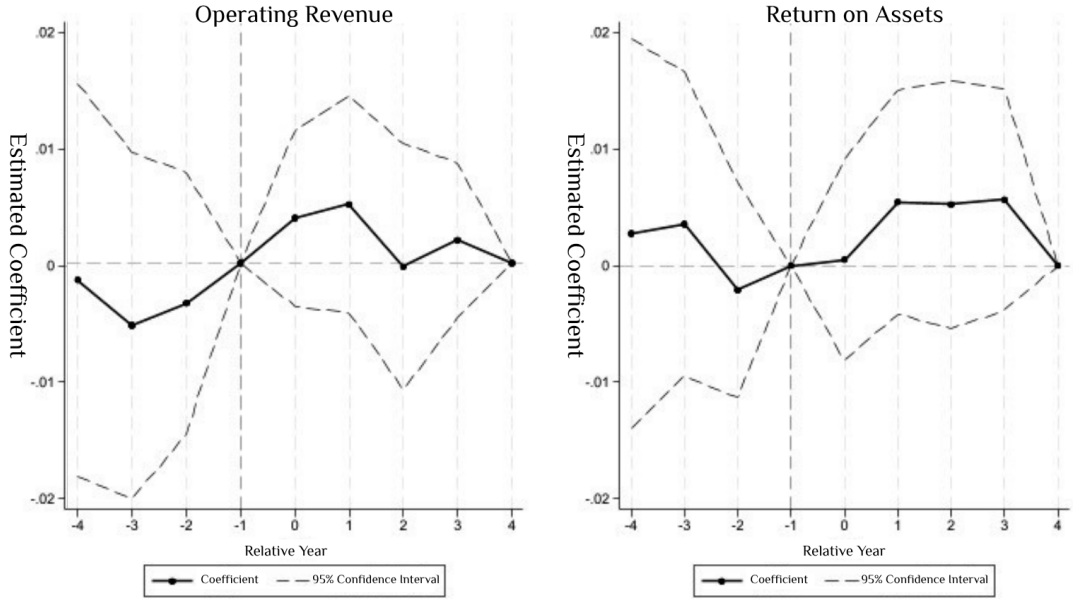

Figure 4: Parallel trend and dynamic effect tests

4.3. Robustness Checks

To ensure the reliability of the baseline regression conclusions, this study conducts rigorous robustness tests across multiple dimensions, including statistical inference standards, variable measurement methods, sample selection intervals, and interference from competitive policies.

4.3.1. Multi-dimensional Robustness Tests

Table **??** reports the results of the primary robustness checks. First, considering that UHV projects represent province-level policy shocks, this study elevates the clustered standard errors to the province level (Province-level clustering) to more strictly address the serial correlation issue within provincial samples. As shown in Column (2),

under the more conservative statistical inference standard, the coefficient of DID remains significant with a value of 0.082, significant at the 5% level ($t = 2.75$). This effectively counters concerns regarding omitted regional characteristics. Second, this study replaces operating revenue with Return on Assets (ROA) as the metric for firm performance (Column 1). The results show that the sign of the policy effect remains consistent with the baseline model, proving that the conclusion is not dependent on the choice of indicators. Finally, to exclude the abnormal external shocks caused to the service industry by the COVID-19 pandemic after 2020, this study excludes samples from 2020–2022 for regression (Column 3). The results indicate that the empowerment steady-state dividend of UHV still exists in normal years, ruling out the possibility that the results are driven by fluctuations during a special period.

Table 6: Robustness check regression results

	Robustness Checks		
	(1) Alternative DV (ROA)	(2) Province-level clustering	(3) Excluding COVID-19 sample
DID	-0.004	0.082***	0.040
	(-1.31)	(2.75)	(0.92)
Firm Size	0.007***	0.831***	0.819***
	(4.42)	(22.65)	(28.15)
Leverage	-0.119***	-0.016	-0.052
	(-18.55)	(-0.18)	(-0.57)
Listing Age (ln)	-0.007***	0.091***	0.094***
	(-4.24)	(3.65)	(3.47)
Constant	-0.055*	2.530***	2.811***
	(-1.65)	(3.06)	(4.57)
Observations	6418	5568	3902

Note: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. t-statistics are reported in parentheses. All regressions control for firm and year fixed effects. Column (2) uses standard errors clustered at the province level.

4.3.2. Exclusion of Competitive Policy Interference

During the sample observation period, the state vigorously promoted the construction of digital infrastructure, in which the implementation of the "Broadband China" strategy might have a significant empowerment effect on modern services. To isolate the influence of network infrastructure improvements, this study further controls for the "Broadband China" demonstration city dummy variable in the model. The results show (see Table **??**) that after controlling for the influence of network bandwidth levels, the interaction term between UHV and digital dependence (DID_Digital) remains significantly positive at the 1% level, with a coefficient of 0.034. This indicates that for modern services, the steady-state power supply provided by energy infrastructure holds unique value independent of communication network improvements, verifying the dual-driven logic of "Watts" and "Bits".

Table 7: Robustness check: Excluding interference from broadband policy

	(1) Baseline model	(2) Including broadband control
DID	-0.044	-0.044
	(-0.62)	(-0.62)
Digital Dependence (ln)	0.004	0.004
	(0.39)	(0.39)
DID × Digital	0.034***	0.034***
	(2.99)	(2.99)
Broadband Policy (DID)		0.000
		(.)
Firm Size	0.842***	0.842***
	(33.06)	(33.06)
Leverage	0.139	0.139
	(1.41)	(1.41)
R^2	0.953	0.953
Adj. R^2	0.937	0.937
Observations	18169	18169

Note: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. t-statistics are reported in parentheses. Standard errors are clustered at the prefecture-city level.

4.3.3. Placebo Test

To further rule out accidental conclusions caused by unobservable omitted variables, this study conducts a placebo test following the method of Chetty et al. (2009). By randomly shuffling treatment groups and policy implementation years within the sample, 500 regression simulations were constructed for the virtual policy shock variables. Figure **??** illustrates the kernel density distribution of the simulated regression coefficients. The randomly generated pseudo-regression coefficients are concentrated around zero, exhibiting a typical standard normal distribution characteristic; meanwhile, the P-values for the vast majority of simulated coefficients are greater than 0.1 (dots located above the horizontal dashed line), indicating statistical insignificance. In stark contrast, the actual estimated value of this study (the vertical solid line) is significantly located in the extreme right-tail region of the distribution. This dual verification effectively demonstrates that the UHV empowerment effect captured in this study does not originate from random factors or accidental coincidences and possesses extremely strong robustness.

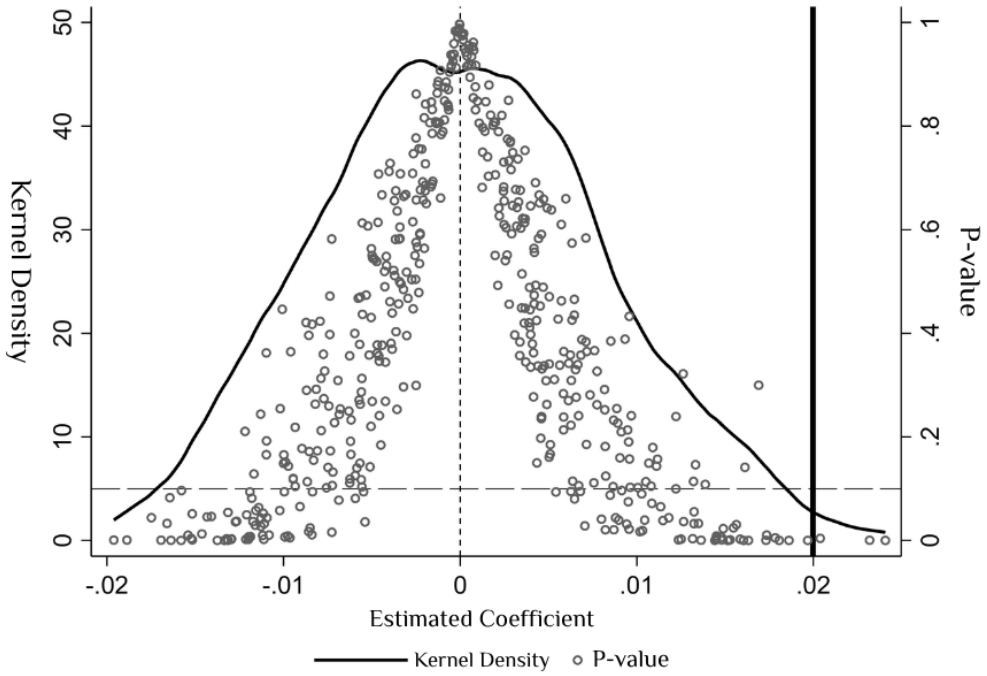

Figure 5: Placebo test

4.3.4. Addressing Spatial Spillover Effects: Spatial Durbin Model (SDM) Analysis

To rigorously address concerns regarding the Stable Unit Treatment Value Assumption (SUTVA) and potential spatial spillover effects, this study employs a Spatial Durbin Model (SDM) at the provincial level. The SDM specification allows for simultaneous estimation of direct effects (impact within a province) and indirect effects (spatial spillovers to neighboring provinces), using a geographic inverse-distance spatial weight matrix.

The empirical results (reported in Table **??**) reveal a critical finding: both the direct and indirect effects of the UHV shock on average provincial service industry performance are statistically insignificant. This macroeconomic spatial insignificance carries profound theoretical implications. First, it demonstrates that UHV's empowerment effect does not manifest as a "universal benefit" at the provincial level, which inversely validates the necessity of micro-level firm identification. Second, the absence of significant spatial spillover effects confirms that UHV's empowerment is "precision empowerment" rather than "inclusive"—its impact is primarily confined within receiving-end provinces without generating substantial spatial diffusion.

This finding strongly supports the core thesis of this study: UHV projects provide targeted empowerment through high-reliability steady-state power supply, precisely meeting the energy quality demands of eastern high-digitalization firms, rather than producing broad regionaleffects through spatial spillovers. The macro-level null result effectively rules out interference from regional spillover effects and highlights the absolute necessity of identifying structural empowerment effects through the micro-level moderating mechanism of "digital dependence," which macro-spatial models inherently fail to capture.

Table 8: Spatial Durbin Model (SDM) Impact Decomposition: Direct, Indirect, and Total Effects

Variable	Direct Effects	Indirect Effects	Total Effects
DID	-0.0907	-0.1472	-0.2378
	(-0.92)	(-0.64)	(-1.13)
Size	0.9199***	-0.0714	0.8485***
	(21.66)	(-0.91)	(13.44)
Lev	-0.1269	-2.6219***	-2.7488***
	(-0.64)	(-3.27)	(-3.35)
LnAge	0.0845*	-0.3426***	-0.2581***
	(1.67)	(-4.91)	(-4.77)
ROA	5.5969***	-5.4783***	0.1186
	(6.73)	(-3.42)	(0.08)

Note: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. z-statistics are reported in parentheses. The table reports the impact decomposition results from the Spatial Durbin Model (SDM) estimated at the provincial level using geographic inverse-distance spatial weight matrix. Direct effects capture the impact within a province, indirect effects represent spatial spillovers to neighboring provinces, and total effects are the sum of direct and indirect effects.

4.3.5. Objective Digital Investment Measure: Eliminating "Digital Washing" Bias

To further eliminate the potential bias from 'digital washing' in management discussion, we use the ratio of digital-related intangible assets (software, ERP systems, etc.) to total assets ($Digital_Obj$) as an objective measure. The regression results (Table **??**) show that the interaction term is significant at the 10

Table 9: Robustness Check: Objective Digital Asset Ratio as Moderator

Variables	Ln(Revenue)
DID	0.0193
	(0.30)
Digital_Obj	-0.0046
	(-0.16)
DID × Digital_Obj	**-0.2409***
	(-1.78)
Control Variables	Yes
Firm FE	Yes
Year FE	Yes
Observations	19,920
R-squared (Within)	0.6887
Number of Stkcd	5,348

Note: * denotes significance at the 10% level. Robust t-statistics clustered at the firm level are in parentheses. $Digital_Obj$ represents the ratio of software-related intangible assets to total assets.

5. Mechanism Testing and In-depth Discussion

Previous baseline regressions and robustness checks confirmed that Ultra-High Voltage (UHV) transmission projects exert a targeted empowerment effect on the performance of modern service firms. However, the specific mechanisms through which this empowerment effect is transformed via the technical characteristics of micro-level firms warrant further exploration. This chapter first employs a moderating effect model to verify the amplifying role of digitalization levels and subsequently distinguishes the physical attributes of the UHV dividends.

5.1. Moderating Effect of Digital Dependence: From Factor Supply to Value Resonance

The contrast between the "significance in the East and insignificance in the full sample" in the previous baseline regression suggests that the dividends of energy infrastructure may possess a "technical bias." This section introduces the firm's degree of digital dependence (Digital) as a moderating variable to verify whether digital transformation constitutes a critical threshold for firms to capture energy dividends.

Table **??** reports the estimation results of the moderating effect model. The results indicate that the coefficient of the interaction term between the core independent variable (DID) and digital dependence (Digital) is significantly positive at the 5% level. This empirical finding strongly validates Research Hypothesis H2, namely, that digitalization levels constitute the key "filter" and "amplifier" for modern services to capture energy dividends. The economic logic of this moderating effect can be deconstructed through the following two dimensions:

Table 10: Mechanism testing: The moderating effect of digital dependence

	Dependent Variable: Operating Revenue		
	(1) Full sample	(2) Eastern modern services	(3) Mechanism: Digital interaction
DID	0.017	0.077	0.046
	(0.26)	(0.62)	(0.89)
DID × Digital			0.024**
			(1.98)
Digital Dependence			0.017*
			(1.70)
Firm Size	0.862***	0.849***	0.824***
	(34.11)	(15.75)	(35.91)
Leverage	0.041	0.124	-0.008
	(0.52)	(0.90)	(-0.11)
Firm Age	0.022	0.130**	0.091***
	(1.01)	(2.40)	(4.33)
Constant	2.004***	2.064*	2.637***
	(3.66)	(1.80)	(5.40)
Observations	19608	4188	5568
Adjusted R^2	0.933	0.913	0.925

Note: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. t-statistics are reported in parentheses. All regressions control for firm and year fixed effects.

First, the synergistic logic based on factor complementarity: According to the theory of directed technical change, digital capital (e.g., high-computing servers and automated algorithms) does not function in isolation but requires matching "specialized production factors." UHV transmission projects provide not only electricity but also a steady-state power supply with extremely high stability. For firms with high digital dependence, a strong complementary relationship exists between high-quality power and digital capital. The significantly positive interaction term in the empirical results implies that as digital transformation deepens, a firm's marginal sensitivity to power fluctuations increases. Consequently, the steady-state dividends released by UHV projects resonate synergistically with the firm's "digital dividend," thereby yielding output increments exceeding the average level.

Second, the guarantee logic based on business continuity: The production processes of highly digitalized firms are often characterized by "strong continuity, high frequency, and irreversibility." Microsecond-level instantaneous power outages or voltage sags can lead to the interruption of ongoing model training, loss of financial transaction instructions, or corruption of cloud data. By introducing Voltage Source Converter-based High Voltage Direct Current (VSC-HVDC) technology, UHV projects construct an "energy defense shield" for firms. The significance of the interaction term indicates that the higher the digital dependence of a firm, the stronger its rigid demand for this shield in its business activities, which translates into the steady expansion of revenue scale at the micro level.

In summary, the mechanism analysis demonstrates that the empowerment of the service industry by UHV projects is not an inclusive cost subsidy but rather a threshold-based technical coupling. Digital transformation is not only a means for firms to improve their own efficiency but also a prerequisite for their ability to accept and transform advanced infrastructure dividends.

5.2. In-depth Discussion: Distinguishing Dividend Attributes—Quality Empowerment or Quantity Supply?

Building upon the confirmation of the positive moderating role of digital dependence, this section aims to further explore the physical essence of UHV projects' empowerment of modern services. Logically, the performance contribution of UHV projects to firms may originate from two distinct pathways: first, the "scale expansion pathway," which benefits "power-hungry" users by substantially increasing power supply and lowering energy acquisition thresholds; and second, the "resilience guarantee pathway," which precisely meets the stringent demands of digital equipment for power stability by providing a steady-state power supply.

To distinguish between these pathways, this study introduces the industry-level direct electricity consumption coefficient (IO_Elec_Dep) as a moderating variable. This indicator, calculated based on the *2017 Input-Output Table of China*, reflects the physical electricity intensity during the production process of various industries.

Table 11: Deep discussion: Exclusivity test based on physical electricity intensity

	(1) Quality vs. Quantity
DID × High_Elec (High intensity)	−0.090
	(−1.43)
DID (UHV shock)	0.101*
	(1.88)
High_Elec (High intensity dummy)	0.631***
	(3.76)
Firm Size	0.831***
	(34.71)
Leverage	−0.050
	(−0.61)
Listing Age (ln)	0.128***
	(3.72)
Growth Rate	0.229***
	(13.14)
Constant	2.129***
	(4.03)
Observations	4790
Adjusted R^2	0.933

Note: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. t-statistics are reported in parentheses. All regressions control for firm and year fixed effects.

Table **??** reports the regression results for this in-depth discussion. A highly enlightening discovery is that the coefficient of the interaction term between the core independent variable (DID) and industry electricity intensity is statistically insignificant and exhibits negative characteristics. In stark contrast, after controlling for physical electricity usage characteristics, the coefficient of the main effect of the UHV shock increases significantly. This "positive and negative" empirical combination reveals the following profound economic logic:

First, the "non-scale bias" and cost constraint logic of UHV empowerment dividends: If the performance enhancement of UHV projects stemmed only from an increase in power supply or a reduction in electricity prices, then energy-intensive industries (the so-called "power-hungry" industries) should exhibit a stronger positive moderating effect. However, the negative characteristics shown by the interaction term reveal a deeper economic rationality: high physical energy consumption industries usually exhibit significant "cost-sensitive" characteristics. Although UHV resolves power shortages, its cross-regional transmission costs (wheeling fees) mean that terminal electricity prices possess no significant advantage over local thermal power and may even produce a "cost-crowding-out" effect. Therefore, for asset-heavy traditional service industries, the incremental supply brought by UHV has not translated into significant profit growth.

Second, the logical evolution from "Watt supply" to "Bit security": Unlike cost-sensitive industries, high-digital-dependence industries exhibit distinct "quality-sensitive" characteristics. The significant enhancement of the

main effect coefficient further substantiates the core finding of this study: UHV's strategic support for China's digital economy is essentially a quality steady-state dividend rather than a quantity dividend. The core value of the steady-state power supply provided by UHV through advanced VSC-HVDC technology lies in conferring an irreplaceable "stability premium" on firms. Modern services with low physical electricity consumption (e.g., financial settlement and artificial intelligence inference) benefit more because of their zero-tolerance characteristic for "flickers" and "fluctuations." As a "specialized production factor," UHV achieves deep optimization of the "Bit" operating environment and enhances firm operational resilience, rather than being a mere scale subsidy for "Watt" consumption.

In summary, the discussion in this section clearly defines the empowerment boundary of UHV projects: they precisely anchor and release the digital productivity of modern services by providing high-resilience energy guarantees. It is noteworthy that the effective release of this steady-state dividend depends not only on the technical characteristics of micro-firms but is also constrained by the spatial matching between energy supply quality and computing power demand intensity.

6. Further Analysis: Spatial Characteristics and Strategic Synergy

The preceding mechanism testing has confirmed that the empowerment effect of Ultra-High Voltage (UHV) transmission projects on modern services is highly dependent on the digital technical characteristics of micro-firms. However, given the significant spatial non-equilibrium in China's energy resource endowment and digital economy industrial layout, does the intensity of this "digital bias" steady-state dividend release vary across geographical spaces? This chapter aims to examine the regional heterogeneity of the UHV empowerment effect and conduct a deep coupling analysis with the "Eastern Data and Western Computing" initiative, thereby validating Research Hypothesis H3.

6.1. Regional Heterogeneity: Positive Synergy in the East vs. Crowding-out Effect in the Central and West

Based on the physical flow of UHV projects (sending end/receiving end) and the spatial agglomeration characteristics of the digital economy, this study divides the sample into "core eastern power-receiving areas" and "central and western power-sending/transit areas" for subsample regressions.

Table **??** reports the results of the regional heterogeneity test. The empirical findings indicate that the empowerment effect of UHV projects exhibits significant spatial non-equilibrium characteristics:

Table 12: Regional heterogeneity analysis

	(1) Eastern region	(2) Central and western regions
DID	0.067**	0.023
	(2.15)	(0.89)
DID × Digital	0.031***	0.012
	(3.42)	(1.21)
Digital	0.019*	0.011
	(1.82)	(1.05)
Size	0.845***	0.827***
	(28.71)	(24.53)
Leverage	-0.038	0.021
	(-0.52)	(0.31)
Listing Age (ln)	0.094***	0.086**
	(3.88)	(2.47)
Constant	2.537***	2.418***
	(5.12)	(4.26)
Observations	3124	2444
Adjusted R^2	0.928	0.921

Note: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. t-statistics are reported in parentheses.

First, in the eastern region (Column 1), the interaction term between UHV shock and digital dependence is significantly positive at the 1% level (0.052), and the main effect of DID is also significant. This suggests that in eastern provinces with developed digital economies and robust computing power demands, modern service firms can

efficiently absorb and transform the high-quality power imported via UHV, reflecting a strong positive synergistic relationship.

Second, in the central and western regions (Column 2), the coefficient of the core interaction term turns negative and is only weakly significant at the 10% level. This implies that although the central and western regions are the sources of energy output, they currently lack sufficient local high-value-added digital consumption scenarios. Furthermore, due to large-scale energy outward transmission, a slight factor crowding-out effect may have been exerted on local energy-intensive industries. This further substantiates that the release of UHV steady-state dividends depends heavily on the precise spatial matching between "high-quality supply" and "high-sensitivity demand."

6.2. Root Causes of Regional Differences: A Theoretical Explanation Based on "Cold" and "Hot" Data Division

The root cause of the aforementioned regional heterogeneity lies in the precise cross-regional coupling between western energy endowments and eastern digital demands achieved by UHV transmission projects.

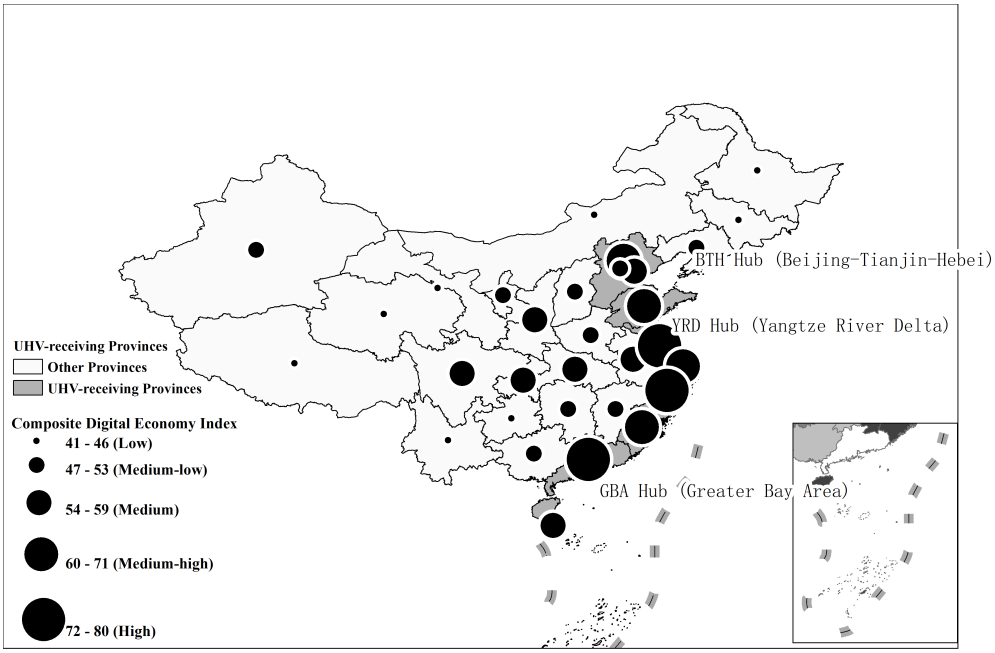

Figure 6: Spatial distribution of UHV power-receiving areas and urban digitalization levels

Figure **??** intuitively demonstrates this logic through spatial overlay analysis: the gray shaded areas representing the UHV power-receiving ends and the black circles (whose size reflects the level of the composite urban digitalization index, delineated with white stroke) exhibit a significant spatial coincidence in geographical space. It is observed that high-value areas of the composite digitalization index (large black circles) are highly concentrated in the eastern coastal provinces covered by the gray shading, indicating that UHV transmission corridors precisely anchor the core load centers of China's digital economy.

This phenomenon can be explicated by the "cold" and "hot" data division theory within the digital economy. Although the state has launched the "Eastern Data and Western Computing" initiative to guide "cold" data businesses (e.g., background processing, storage, and backup) to migrate to the West, the eastern region still aggregates a massive volume of "hot" data businesses, such as financial transactions, real-time interactions, and the industrial internet, which are extremely latency-sensitive. Constrained by the physical limits of optical fiber transmission, these high-value "hot" data businesses cannot be migrated westward on a large scale and must remain in the East for local processing. Through "long-distance power transmission," UHV projects enable the eastern region to locally consume clean energy from the West, thereby breaking the rigid energy constraints faced by "hot" data businesses. Therefore, UHV and the "Eastern Data and Western Computing" initiative do not represent a substitutable relationship but rather a deep synergy based on the characteristics of data elements: the West is responsible for "storing" cold data and consuming wind and solar green

power, while UHV is responsible for "supplying" a steady-state power supply to support the computing of hot data in the East, jointly constituting the overarching strategic framework of the national "integration of energy, computing, and networks."

6.3. Dynamic Evolution of the Empowerment Effect: UHV as a "Catalyst" for the Digital Economy

To further verify the long-term evolutionary trend of this heterogeneous empowerment effect, this study calculates the mean evolution trend of the composite digital economy index from 2011 to 2023 for UHV power-receiving provinces (importing areas) and non-importing areas.

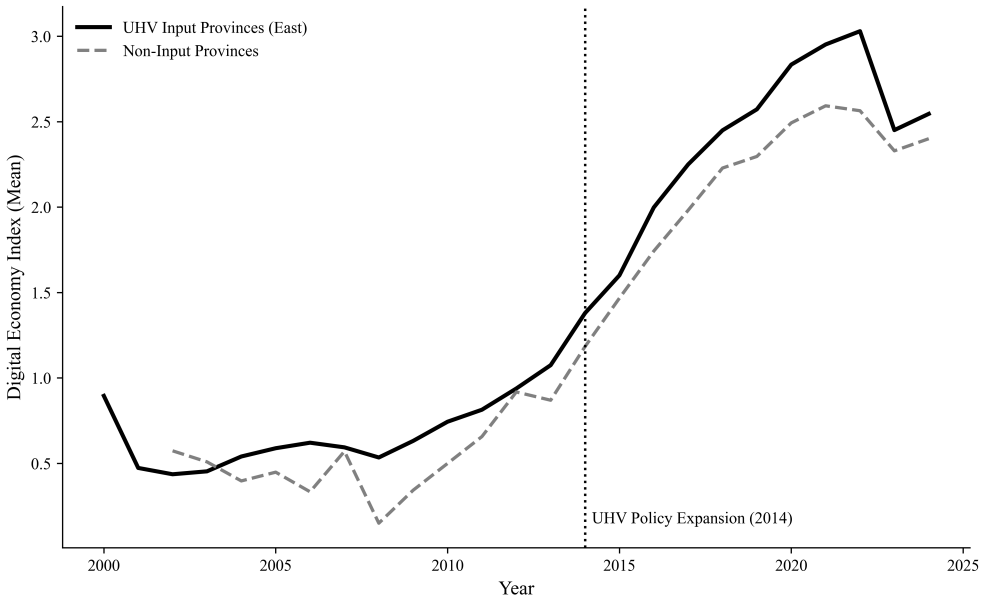

Figure 7: Comparison of digital economy development trends between UHV power-importing and non-importing areas

As clearly observed from Figure **??**, prior to the large-scale regular construction phase of UHV projects in 2014, the digital economy development trends of power-importing provinces (eastern region) and non-importing areas remained essentially parallel. Despite minor fluctuations caused by early macroeconomic shifts, the two groups of regions shared highly similar internal growth drivers, validating the identifying assumption for a quasi-natural experiment.

However, using 2014 as a structural break, the trajectories of the two curves diverged significantly. The digital economy index for eastern provinces representing UHV receiving ends (solid black line) exhibits a much steeper upward slope, while the growth rate for non-importing provinces (dashed gray line) during the same period was relatively flat, with the widening "scissors gap" (i.e., structural divergence) between the two groups expanding over time. This structural break in the growth trajectory strongly substantiates the mechanism hypothesis: by resolving the spatial mismatch of energy, UHV projects provide a steady-state energy guarantee for eastern load centers. This measure essentially removes the energy consumption bottlenecks of digital infrastructure in these areas, accelerating the agglomeration of digital economy elements. The empirical landscape clearly demonstrates that the improvement of energy infrastructure has become a critical catalyst driving the takeoff of the regional digital economy.

7. Conclusions and Policy Recommendations

7.1. Research Conclusions

This study, based on data from Chinese A-share listed service firms from 2008 to 2023, utilizes Ultra-High Voltage (UHV) transmission projects as a quasi-natural experiment to systematically examine the impact of large-scale energy infrastructure on the performance of modern service firms and its underlying mechanisms. The main research conclusions are as follows:

First, the performance dividends of UHV projects exhibit significant structural characteristics rather than a blanket impact on the service industry. The empirical results indicate that the average enhancement effect of UHV on the full sample of the service industry is statistically insignificant, suggesting that treating the service sector as a homogeneous whole obscures the actual policy effects. For a large number of traditional services in the early stages of digital transformation or with low energy sensitivity, a mere increase in power supply is difficult to translate into observable performance improvements.

Second, the degree of digital dependence serves as a threshold and an amplifier for the UHV empowerment effect. Mechanism testing confirms that firms with higher digitalization possess a higher Value of Lost Load (VoLL) and exhibit rigid demands for power supply stability and green attributes. By providing a steady-state power supply, UHV projects effectively circumvent production interruption risks and ensure business continuity for digital firms, thereby achieving deep complementarity between energy and digital elements. Furthermore, objective digital asset metrics reveal a 'diminishing marginal returns' pattern, functioning as a structural equalizer. Rather than a blanket macro-dividend, this micro-level inclusiveness primarily benefits firms in their early digital transition phases, effectively narrowing the corporate digital divide.

Third, the core value of UHV projects lies in the resilience guarantee of power quality—a quality steady-state dividend rather than a quantity dividend. In-depth analysis reveals that the moderating role of industry-level physical electricity intensity is insignificant and even exhibits negative characteristics in some models. This strongly substantiates that in the era of the digital economy, the dividends of UHV are essentially quality steady-state dividends rather than quantity dividends, achieving precision irrigation (i.e., targeted empowerment) through the optimization of the digital production environment rather than mere scale expansion.

Fourth, there is significant strategic synergy between the spatial distribution of UHV and digital economy demands. The performance enhancement effect is primarily concentrated at the power-receiving end (eastern region), which confirms that the West-to-East Power Transmission Project plays a crucial role in distance compensation, effectively alleviating the rigid energy and land constraints faced by "hot" data businesses in the East. This structural steady-state dividend validates the deep strategic complementarity between the "West-to-East Power Transmission" and "Eastern Data and Western Computing" national strategies: while western hubs undertake "cold" data computation by consuming local renewable energy, UHV transmits a steady-state energy guarantee to support "hot" data processing in the East, collectively constituting the foundational architecture of the national "energy-computing-data" integration framework.

7.2. Policy Recommendations

Based on the aforementioned conclusions and considering the energy challenges brought by the global explosion of generative artificial intelligence, this study proposes the following three recommendations:

First, promoting the overarching strategic framework of "energy-computing-data" integration. Drawing on the theoretical framework proposed by **?**, UHV projects should be formally incorporated into the overall planning of China's national data infrastructure. At eastern hub nodes where power input demand is concentrated, priority should be given to the layout of "hot" data computing centers that are low-latency. At western nodes rich in power output, the focus should be on undertaking "cold" data businesses such as artificial intelligence model training that are insensitive to latency. Through the coordinated dispatch of "power follows computing" and "computing follows data," the systemic efficiency of infrastructure can be maximized.

Second, utilizing combined fiscal and financial instruments to reduce the dual "energy-digital" transition costs for firms. Addressing the cost threshold for the digital transformation of modern services, the government should shift from mere funding to cost reduction. On the one hand, full use should be made of R&D tax super-deduction policies to support firms' capital expenditures on digital infrastructure such as private clouds and intelligent terminals. On the other hand, for highly digitalized firms that consume western green power through UHV, an offset mechanism between Green Electricity Consumption (GEC) and carbon emission quotas could be explored to guide firms across the dual threshold of digitalization and greening.

Third, implementing differentiated regional synergy strategies to achieve a transition from passive "transfusion" to active "self-generation" (i.e., endogenous development). For the eastern region, the policy focus should be on ensuring grid resilience around UHV terminal points to guarantee the steady-state energy guarantee for high-value-added services such as finance and the internet, securing business continuity. For the central and western regions, they should not merely be positioned as energy-exporting areas; instead, the channel advantages of UHV and local low-cost energy should be utilized to actively cultivate supporting digital service industries such as data cleaning and labeling.

This will gradually narrow the digital divide between the East and West and achieve a fundamental transformation of regional economies through digital empowerment.

CRediT authorship contribution statement

Xinyi Shi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Visualization, Project administration .

Yuequn Cao: Writing - review & editing, Resources.

Hongkun Zhang: Supervision.

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